
SPARSITY AND SUPERPOSITION IN MIXTURE OF EXPERTS

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ABSTRACT

Mixture of Experts (MoE) architectures now underpin the scaling of contemporary large language models, yet the mechanistic principles that separate them from dense networks remain largely uncharacterized. Prior work established that dense models exploit *superposition*—encoding more features than they possess dimensions—and that the extent of this superposition is jointly governed by feature sparsity and feature importance. We show that this dense-model lens does not carry over to MoEs: neither feature sparsity nor feature importance induces the discontinuous phase changes that characterize dense representations. The governing variable is instead *network sparsity*—the ratio of active to total experts—which determines the representational strategy an MoE adopts. To render this behavior measurable, we introduce expert-resolved metrics that quantify superposition at the granularity of individual experts, and we establish a systematic, monotonic relationship: models with greater network sparsity exhibit correspondingly greater *monosemanticity*. Building on this result, we advance a definition of expert specialization grounded in monosemantic feature representation rather than in load balancing, and we demonstrate that experts organize around coherent feature combinations whenever initialization directs them toward those features—thereby recovering load balancing as a consequence of specialization rather than its definition. Taken together, these findings establish that network sparsity renders MoEs more interpretable at only a modest, and steadily narrowing, cost in reconstruction performance, directly challenging the prevailing assumption that interpretability and capability stand in fundamental tension.

1 INTRODUCTION

Mixture of Experts (MoEs) have become the dominant architecture in state-of-the-art language models, including Qwen3, Mixtral, and Gemini (Yang et al., 2025a; Jiang et al., 2024; Google DeepMind, 2025), principally because conditional computation delivers substantial simultaneous gains in efficiency and performance (Shazeer et al., 2017; Fedus et al., 2022). A rich body of subsequent work has refined this design space—improving routing through mechanisms such as Expert-Choice routing and stabilizing training and transfer through approaches such as ST-MoE (Zhou et al., 2022; Zoph et al., 2022)—while complementary theoretical and empirical results have shown that learnable routers recover latent cluster structure in the data, furnishing a first account of why experts specialize (Dikkala et al., 2023). Despite this widespread adoption and architectural maturity, however, the internal representations of MoEs remain poorly understood from a mechanistic-interpretability standpoint. It is precisely this gap that the present study sets out to close.

A parallel line of interpretability-oriented work has sought to render expert behavior more transparent. Yang et al. (2025b) propose MoE-X, which couples sparsity-aware routing with wide, ReLU-based experts to suppress polysemanticity, while Park et al. (2025) introduce MONET, scaling the expert count so that capabilities can be edited through targeted expert activation. These contributions operate principally at the level of architectural intervention: they impose interpretability from without, and consequently leave unresolved the more fundamental questions of how MoEs represent features intrinsically, how the expert

structure shapes superposition, and whether experts specialize of their own accord in the absence of additional regularization. Surveying the field, Mu & Lin (2025) identify exactly this mechanistic understanding as a central open challenge. Our study addresses it directly, characterizing the intrinsic representational geometry of MoEs rather than engineering interpretability onto them.

A fundamental obstacle to interpreting neural networks is the phenomenon of superposition: the regime in which a model represents more features than it has dimensions. Superposition lets a network pack many sparse features into a smaller set of neurons, but it does so at a direct interpretive cost, rendering individual neurons polysemantic and thereby resistant to clean, feature-level analysis. Understanding when a model resorts to superposition—and what architectural factors relieve the pressure to do so—is thus a prerequisite for mechanistic interpretability.

MoE architectures introduce a new axis to this problem: network sparsity. Whereas dense models activate every neuron irrespective of the input, an MoE activates only a fraction of its total parameters on any given forward pass (Shazeer et al., 2017). This distinction is representationally decisive. A dense model is compelled to exploit feature sparsity by packing many sparse features into shared neurons; an MoE, by contrast, can afford to be selective, dedicating entire experts to specific feature combinations rather than compressing them into a common basis. Network sparsity therefore endows MoEs with a representational degree of freedom that dense models simply lack, and it is the consequences of this degree of freedom that we characterize—showing that it, and not feature sparsity, is what governs whether an MoE represents its features in superposition or monosemantically.

We organize our investigation around three questions: (1) whether MoEs exhibit less superposition than their dense counterparts; (2) whether the superposition of a given feature in an MoE expert undergoes the discrete phase change—across the feature’s relative importance and the overall feature sparsity—that is observed in dense models; and (3) whether expert specialization is more faithfully understood through the lens of feature representation than through load balancing alone.

We address these questions using controlled toy models that extend the framework of Elhage et al. (2022) to the MoE setting, a design that isolates the effect of network sparsity while holding parameter budgets fixed. This controlled comparison lets us attribute every observed difference to the architecture itself rather than to a difference in capacity, and it yields three principal contributions: (1) we establish that, unlike dense models, MoEs do not undergo sharp phase changes, exhibiting instead increasingly continuous representational transitions as network sparsity increases; (2) we demonstrate that MoEs consistently attain greater monosemanticity—equivalently, less superposition—than dense models matched on both active and total parameters, with individual experts representing features more cleanly and confining interference to within-expert blocks; and (3) we introduce an interpretability-focused definition of expert specialization grounded in monosemantic feature representation, and we show that experts organize around coherent feature combinations rather than the arbitrary load balancing assumed by conventional accounts—so that load balancing emerges as a consequence of specialization rather than its defining criterion.

2 RELATED WORK

Superposition and Feature Representations. The Linear Representation Hypothesis holds that networks encode concepts as directions in activation space (Park et al., 2024), yet the number of interpretable features a network must represent routinely exceeds its available dimensions. Elhage et al. (2022) formalized the resolution of this tension as superposition, establishing that dense models pack non-orthogonal features so as to maximize representational capacity at the cost of polysemanticity. Although superposition is provably efficient as a capacity strategy (Scherlis et al., 2025), it forces reliance on elaborate post-hoc disentanglement methods—most prominently Sparse Autoencoders—to recover monosemantic features (Bricken et al., 2023). More recent work characterizes how data correlations shape superposition (Prieto et al., 2025) and how interference patterns arise and may be mitigated (Gurnee et al., 2023). We extend this line of inquiry to MoE architectures and identify the

controlling variable: it is network sparsity, rather than feature sparsity alone, that governs the representational strategy a model adopts.

Interpretability of MoEs. Although MoEs have become the standard vehicle for scaling large language models (Shazeer et al., 2017; Fedus et al., 2022; Jiang et al., 2024), mechanistic understanding of their internal representations lags well behind that of dense networks. Existing analyses concentrate on macroscopic behaviors—routing stability (Zoph et al., 2022), expert-choice statistics (Zhou et al., 2022), and latent cluster discovery (Dikkala et al., 2023)—rather than on the geometry of the representations themselves. The more recent interpretability-focused proposals attempt to impose specialization through architectural constraints, whether predefined concept routing (Yang et al., 2025b) or expert counts scaled to match vocabulary sizes (Park et al., 2025); these, however, prioritize capability editing over an account of intrinsic feature geometry. We close this gap by analyzing directly how the expert structure shapes superposition: we establish that MoEs exhibit systematically greater monosemanticity than dense models matched on parameters, and we advance a feature-based definition of expert specialization that links this geometry to routing behavior—grounding, rather than assuming, the connection between how an expert represents features and when it is invoked.

3 BACKGROUND

A central aim of mechanistic interpretability is to reverse-engineer neural networks, and the standard approach is to decompose their representations into a set of human-interpretable concepts termed ‘features’. These features are conventionally modeled as linear, so that any hidden state h admits the decomposition

$$h = \sum_{i \in F} \alpha_i \vec{f}_i + \vec{b}$$

where $\vec{f}_i$ is the direction associated with feature i , α_i is that feature’s activation strength—the degree to which feature i is present in the input—and F denotes the set of all represented features.

The *superposition hypothesis* holds that a model can represent far more features than it has dimensions, i.e. $|F| > m$ for $h \in \mathbb{R}^m$ (Elhage et al., 2022). Accommodating more features than dimensions within a latent space necessarily forces the feature vectors in F to be packed non-orthogonally. When two features exhibit nonzero interference, $\langle \vec{f}_i, \vec{f}_j \rangle \neq 0$, we say they are in *superposition*. Such superposition is tolerable precisely when features are sparsely active—a defining characteristic of most online text data—but it exacts an interpretive cost: the recovered coefficient α_i then absorbs spurious activations unrelated to the actual presence of feature i in the input.

Monosemanticity is a property of individual neurons: a neuron is monosemantic when its activation corresponds cleanly to a single coefficient α_i —that is, when features are basis-aligned. Orthogonality and basis-alignment are, crucially, distinct properties: when features are orthogonal (and hence not in superposition) but not basis-aligned, neurons remain *polysemantic* even though feature interference is minimal. Because our concern throughout is with reducing superposition rather than with enforcing basis-alignment, we adopt the following convention for brevity: when we describe a model, expert, or feature as “more monosemantic,” we mean precisely that it exhibits less superposition.

4 DEMONSTRATING SUPERPOSITION

MoEs are commonly conceptualized as compositions of dense models, with each expert treated as an independent dense network. Whether experts in fact represent features in the same manner as dense models, however, has never been established. We test this assumption directly, comparing the superposition that arises within MoEs against that of dense models under matched parameter budgets—and we find the assumption to be false:

the two architectures learn comparable numbers of features but organize their interference in fundamentally different ways.

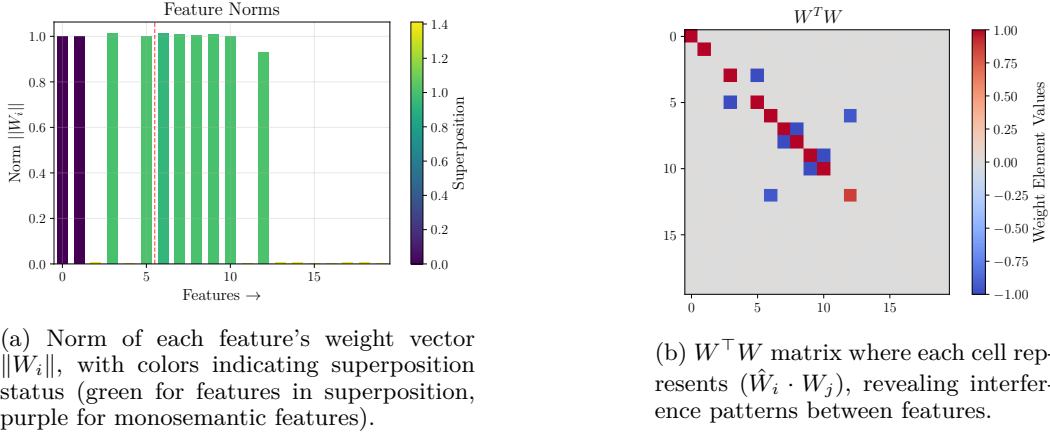


Figure 1: Feature representation and superposition in a dense model with $n = 20$ features and $m = 6$ hidden dimensions, trained under exponentially decaying importance $I = 0.7^i$ and uniform feature density $(1 - S) = 0.1$. The color-encoded superposition score is $\sum_j (\hat{W}_i \cdot W_j)^2$, the total projection of all other features onto the direction of feature i ; larger values indicate that feature i shares its dimensions with more of the remaining features. In the dense model, interference is distributed across the entire feature set, with no partition confining it—the signature against which the block-diagonal structure of the MoE (Figure 2) should be read.

4.1 EXPERIMENTAL SETUP

Our objective is to characterize how an MoE projects a high-dimensional input vector $x \in \mathbb{R}^n$ into lower-dimensional expert representations $h \in \mathbb{R}^m$ and subsequently reconstructs it. This construction extends the toy-model framework of Elhage et al. (2022) to the MoE setting, adding the expert count and routing as the additional degrees of freedom that distinguish an MoE from a dense autoencoder.

The input distribution. The input vector x models the activations of an idealized, fully disentangled model in which each dimension x_i corresponds to a distinct, independent feature—that is, a perfectly neuron-aligned, monosemantic representation. We draw each coordinate $x_i \sim U(0, 1)$ subject to a prescribed sparsity $S \in [0, 1)$, so that $P(x_i = 0) = S$ independently across coordinates. Because features do not contribute equally to the reconstruction objective, we assign each feature x_i a scalar importance I_i that weights its term in the loss. To isolate and study the phase transition of a single feature, we further scale the magnitude of the last feature by a factor $r \in \mathbb{R}^+$, yielding the importance vector $I = (1, 1, \dots, r)$. The parameters I_i and r jointly set the signal strength of individual features, affording precise control over each expert’s sensitivity to feature magnitude.

Model Architecture. The MoE comprises E experts, each expert e parameterized by a weight matrix $W^e \in \mathbb{R}^{m \times n}$ and a bias $b^e \in \mathbb{R}^n$. A learned router $g(x) = \text{softmax}(W^r x)$, with $W^r \in \mathbb{R}^{E \times n}$, assigns each input to its top- k experts. Every active expert first projects the input onto a lower-dimensional hidden state $h^e = W^e x$ and then reconstructs it as $\hat{x}^e = \text{ReLU}((W^e)^T h^e + b^e)^1$. The network output is the gate-weighted sum over the active experts, $x' = \sum_{e \in \text{top-}k} w_e \hat{x}^e$, where the w_e are the renormalized gating weights. Setting

¹The output ReLU enforces non-negative reconstructions, consistent with our input distribution, in which $x_i \in [0, 1]$ whenever nonzero. Moreover, the off-diagonal terms of $(W^e)^T W^e$ produce negative interference, which the ReLU suppresses; when features are sparse, this negative interference becomes effectively “free” because it is clipped to zero, which in turn incentivizes configurations with negative off-diagonal terms, such as antipodal feature pairs.

$E = 1$ recovers a dense autoencoder exactly, so the dense model is the $E = 1$ special case of this family and every comparison we report varies only the expert count and routing.

We train each model to minimize an importance-weighted $L2$ reconstruction loss, $\mathcal{L} = \sum_x \sum_i I_i (x_i - x'_i)^2$, in which the per-feature importances I_i scale the corresponding error terms. In the experiments of Section 4, and to prevent expert collapse, we augment this objective with the standard auxiliary load-balancing loss (Fedus et al., 2022), $\mathcal{L}_{\text{aux}} = \alpha N \sum_{e=1}^N f_e P_e$, where N is the total number of experts, f_e is the fraction of batch samples routed to expert e , P_e is the mean gating probability assigned to expert e over the batch, and $\alpha = 0.01$ sets the penalty strength. We deliberately omit this auxiliary loss in Section 4, isolating the MoE’s intrinsic architectural bias for or against superposition uncontaminated by any externally imposed balancing pressure; where load balancing is nonetheless required to prevent expert collapse, we reintroduce it and say so explicitly.

4.2 MEASURING FEATURE CAPACITY

To analyze feature representations across architectures, we quantify two complementary properties of each feature: its *representation strength* and its *interference* with the remaining features. Representation strength is captured by the norm $\|W_i^e\|$ of feature i ’s weight vector within expert e , which measures the extent to which that expert represents the feature: $\|W_i^e\| \approx 1$ indicates that feature i is fully represented in expert e , whereas $\|W_i^e\| \approx 0$ indicates that it is not learned. Interference is read off the Gram matrix $W^\top W$, whose off-diagonal elements report the pairwise interference between feature i and every other feature within expert e .

At equal total parameters ($m_{\text{dense}} = \sum m_{\text{experts}} = 6$), the dense model and the MoE represent a comparable number of features (10 versus 8) with comparable norms (Figures 1a and 2a). The decisive difference lies not in how many features are learned but in the geometry of their interference. Although individual MoE experts exhibit some local superposition—Expert 0 in Figure 2b, for instance—the global interference structure is strictly partitioned: whereas in the dense model (Figure 1b) any feature may interfere with any other, the MoE enforces a block-diagonal structure in which features routed to distinct experts have exactly zero interference. MoEs thus allocate representational capacity by partitioning the feature space, sharply curtailing the global scope of interference rather than merely redistributing it—the mechanism, we will show, by which network sparsity translates into reduced superposition.

Expert Feature Dimensionality. To make precise how an MoE allocates its limited representational capacity differently from a dense model, we measure feature dimensionality—the “fraction of a dimension” that a given feature occupies within a model (Elhage et al., 2022). For feature i , we define its dimensionality in expert e as

$$D_i^e = \frac{\|W_i^e\|^2}{\sum_j (\hat{W}_i^e \cdot W_j^e)^2} \quad (1)$$

The quantity D_i^e is bounded between zero (feature not learned) and one (feature represented monosemantically), so the total representational capacity of an MoE follows naturally as $D = \sum_e \sum_{i=1}^n D_i^e$.

Efficient Packing. A model packs features “efficiently” when their dimensionalities sum to the number of embedding dimensions, $\sum_{i=1}^n D_i^e \approx m$ (Cohen et al., 2014; Scherlis et al., 2025; Elhage et al., 2022); for an MoE this generalizes to $\sum_e \sum_{i=1}^n D_i^e \approx E \cdot m$. Empirically, both dense and MoE models satisfy this dimensionality constraint, which establishes that MoEs pack features exactly as efficiently as dense models at equal total parameters. Packing efficiency is therefore *not* the axis along which the two architectures differ; the distinction lies entirely in how that capacity is distributed—the question we quantify next.

Features per Dimension & Network Sparsity. Because both dense and MoE models pack features efficiently, the number of features per dimension furnishes a direct,

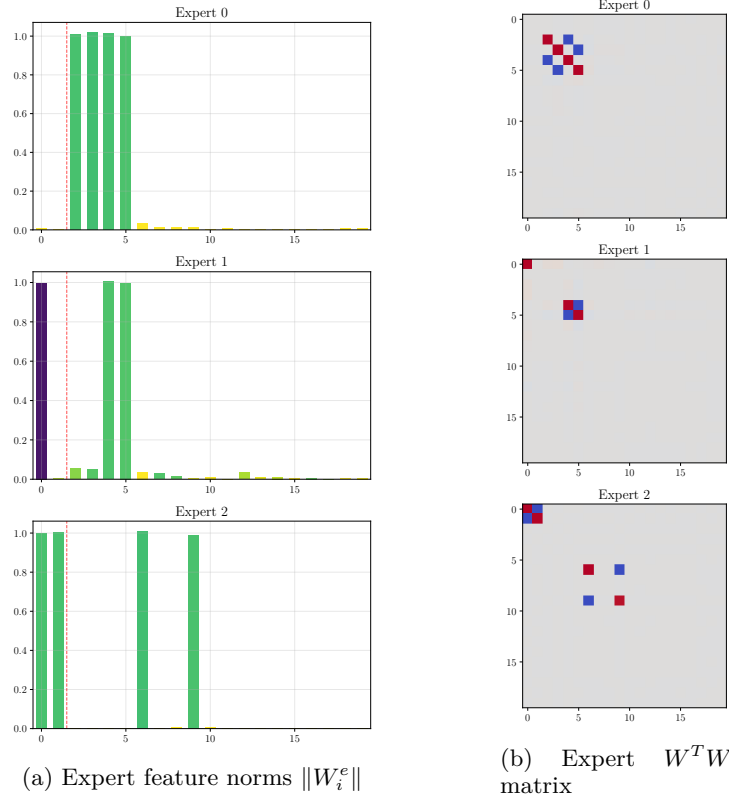


Figure 2: Feature representation and superposition in an MoE with $n = 20$ features, 3 total experts, and $m = 2$ hidden dimensions per expert under $\text{top-}k = 1$ routing, trained under exponentially decaying importance $I = 0.7^i$ and feature density $1 - S = 0.1$. Matched to the dense model of Figure 1 on total parameters, the MoE learns a comparable number of features yet confines their interference to within-expert blocks rather than distributing it across the full feature set—the block-diagonal signature of a partitioned representation, directly contrasting with the fully coupled interference of the dense model.

architecture-agnostic measure of superposition—one that lets us quantify precisely how the expert count modulates superposition across feature sparsities. A features-per-dimension value exceeding one is an unambiguous signature of superposition, since the model then represents more features than it has dimensions. We define features per dimension for an MoE as

$$\frac{1}{k} \sum_{e=1}^E p_e \frac{\|W^e\|_F^2}{m} \quad (2)$$

where $\|W\|_F^2$ is the squared Frobenius norm and p_e is the expected probability that expert e is used across a batch of inputs—equivalently, the average renormalized gating weight after $\text{top-}k$ routing.

At equal total parameters, the dense model registers a consistently higher features-per-dimension value than every MoE (Figure 3), and hence strictly more superposition, at every feature sparsity we test. The dense model exploits superposition to represent more features than it has dimensions ($n_{\text{learned}} > m_{\text{total}}$), whereas the MoE caps its representation at the monosemantic limit ($n_{\text{learned}} \approx m_{\text{total}}$). This behavior is governed by network sparsity: as the total number of experts grows, with total parameters and the ratio k/E held fixed, features per dimension falls monotonically. *The greater the number of experts, the less superposition in the model*—features become progressively more monosemantic as network

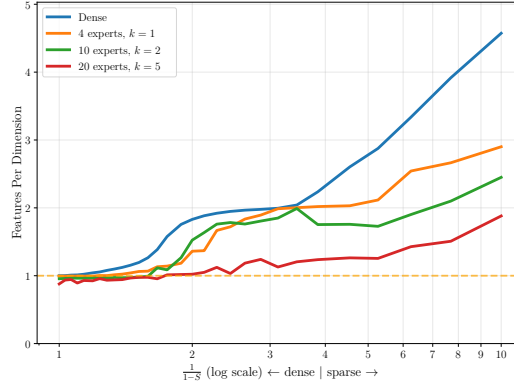


Figure 3: Features per dimension versus inverse feature density ($\frac{1}{1-g}$) for **dense** and **MoE** architectures under uniform feature importance ($I_i = 1.0$). The dense model ($n = 100$, $m = 20$) exhibits the greatest superposition, and superposition falls monotonically as the expert count grows: 4 experts with $m = 5$, $k = 1$ (orange); 10 experts with $m = 2$, $k = 2$ (green); and 20 experts with $m = 1$, $k = 5$ (red). Because all models are matched on total parameters and on the ratio k/E , this ordering isolates the effect of network sparsity alone—no other factor varies across the curves. The dashed line at 1.0 marks the monosemantic limit, below which the model represents no more features than it has dimensions; the MoE curves sit at or below this limit while the dense curve rises above it.

sparsity increases. This interpretability gain, moreover, is not purchased at a prohibitive cost in performance. The dense model’s additional superposition does buy it a consistently lower reconstruction loss than the MoEs, but the gap is small—only ~ 0.03 – 0.08 at any given sparsity (Figure 6, Appendix A.1)—and it closes as the expert count increases: the MoE loss converges toward that of the dense model, so greater monosemanticity is bought under a steadily narrowing performance penalty. Appendix A.2 develops the theoretical intuition underlying this trade-off.

5 PHASE CHANGE

Although MoEs and dense models learn a similar number of features, MoEs distribute those features across experts with substantially less interference. Network sparsity therefore reshapes how features are allocated rather than how many are learned. We now ask how the properties of the input distribution—feature sparsity and feature importance—drive this allocation, and whether they induce the physics-inspired *phase changes* that characterize dense representations. They do not: the discrete transitions that organize dense representations dissolve into continuous ones as network sparsity increases.

A model commands only a finite repertoire of ways to represent any feature: it may ignore the feature, place it in superposition, or represent it monosemantically. A phase change is the observation that discrete boundaries separate these regimes as functions of feature sparsity and relative importance.

Dense toy models exhibit discontinuous ‘phase changes’ between internal feature representations (Elhage et al., 2022): varying the sparsity and relative importance of the input features elicits qualitatively distinct behavior, with greater feature sparsity, for instance, encouraging greater superposition. Analyzing the per-expert phase diagram of an MoE, we find that its experts adopt representational strategies fundamentally different from those of dense models—strategies that lack the sharp boundaries dense models display.

Setup. We adopt the setup of Section 4.1, augmented with the auxiliary load-balancing loss to avert specialization collapse—the failure mode in which experts and routers settle into local minima that leave certain experts entirely unused. We evaluate three configurations, all with a single active expert ($k = 1$): (A) $n = 2$, $m = 1$; (B) $n = 3$, $m = 1$; and (C)

$n = 3, m = 2$. For each, we report the per-expert phase diagram over feature sparsity and last-feature relative importance while sweeping network sparsity, increasing the expert count E up to the number of input feature dimensions n .

Here we hold active parameters fixed rather than total parameters, so that $m_{dense} = km_{moe}$ (disregarding router parameters). This choice yields a genuinely controlled comparison across architectures: were active parameters left free, any observed difference could be ascribed to that budget rather than to the architecture itself. Note that Figure 4.C.1/1 coincides with Figure 4.B.X/2 in active parameters; the latter, however, must encode the same number of input features ($n = 3$) using only a single hidden dimension ($m = 1$) rather than two ($m = 2$), leaving little room to exploit superposition in the service of specialization.

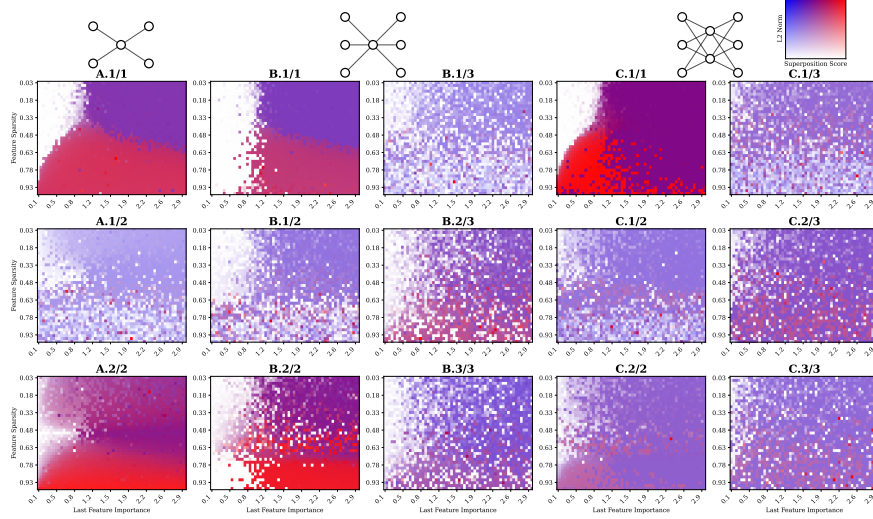


Figure 4: For a given expert and input dimension (feature), we decode how that feature is embedded in the hidden dimension—whether it is ignored (white), represented monosemantically (blue-purple), or placed in superposition (red). Each cell jointly encodes the feature norm ($\|W_n\|^2$) and the superposition score ($\sum_{j < n} (\hat{W}_n \cdot W_j)^2$) of the last feature, swept across feature sparsity $S \in [0.1, 1]$ and relative last-feature importance $I_n \in [0.1, 3]$, where the subscript n denotes the last of n total features. For every cell we train ten models and retain the one attaining the lowest loss; all runs in this section include the load-balancing loss. A low $L2$ norm ($\|W_n\|$) is rendered white to indicate that the model ignores the last feature; otherwise a low superposition score is rendered blue-purple to indicate a monosemantic representation, whereas red indicates that the feature is held in superposition. Cell (i, j) of subfigure X.e/E reports expert e of E total experts, trained on architecture X, for last-feature importance $I_n = i$ and sparsity $S = j$; X.1/1 denotes the corresponding dense model. Reading across each row, the sharp red-to-blue boundary of the dense panel gives way to progressively bluer, more continuously varying panels as the expert count E grows—the visual signature of the phase change dissolving under increasing network sparsity.

Results and Takeaways. In every single-expert (dense) configuration we observe a clear phase change (Figure 4.X.1/1), corroborating Elhage et al. (2022). Increasing the total number of experts systematically dissolves these discrete phase changes. Some experts in the $E = 2$ MoEs echo their dense counterparts (Figure 4.X.2/2) yet transition far more continuously, and in each case one expert becomes markedly more monosemantic, specializing in the feature of highest relative importance. Experts that diverge from the dense pattern uniformly exhibit much lower superposition scores—they are visibly bluer—signaling more monosemantic representations. This is exactly the behavior anticipated in the preceding section: MoEs systematically favor lower superposition than their dense counterparts, and they do so more strongly as network sparsity increases.

In the $n = 2, m = 1$ configuration (Figure 4.A), the dense model declines to represent the last feature under low feature sparsity, whereas the matched MoE preserves that feature far

more readily—precisely because it retains the capacity to do so. With three input dimensions (Figure 4.B) this behavior vanishes: the experts already superimpose the other two features, leaving no room for the third within a single hidden dimension. Architecture B is thus the sole exception, the one in which the superimposing hidden dimension is, in the high-sparsity regime, insufficient to represent all features—and even here no clear phase change emerges, save for the mostly discrete but mixed band at feature sparsity 0.5–0.7 in Figure 4.B.2/2. For $m = 2$, the white region of the dense model (Figure 4.C.1/1)—occupying the mid- to low-sparsity domain where the last feature is relatively less important than the others—reflects that the feature is ignored altogether. As network sparsity increases across the remaining panels of Figure 4.C, the models instead represent the last feature with progressively greater $L2$ magnitude ($\|W_3^1\| < \|W_3^2\| < \|W_3^3\|$). The dimensionality allocated in the low-importance region thus rises monotonically with network sparsity, exactly as Figure 3 predicts—the two experiments converging on the same conclusion from independent measurements.

Across Figures 4.B.2/2, 4.C.1/2, and 4.C.2/2 we observe a window of feature sparsity, roughly 0.48 to 0.7, in which polysemantic, monosemantic, and ignored representations are heavily intermixed. This intermixing marks a middle regime in which polysemantic and monosemantic representations incur comparable loss, so that the MoE cannot consistently commit to the strategies it adopts cleanly in the low- and high-sparsity domains. No analogous regime appears in dense models—direct evidence that MoEs learn qualitatively, and not merely quantitatively, different representational strategies.

Conclusion. All of these experiments use top- $k = 1$ routing, so exactly one expert is active at any time. Even under this most dense-like of settings, markedly different behavior emerges as early as the $E = 2$ case—including when the superimposing hidden dimension has ample capacity to represent every feature. It is therefore misleading to regard an MoE as a mere aggregation of dense models: by restricting each expert to only a subset of the feature domain, the router fundamentally alters both how the experts behave and what they learn. Because $k = 1$ is the setting in which an MoE most closely resembles a dense model, this divergence is likely only more pronounced for $k > 1$.

6 EXPERT SPECIALIZATION

Having established that MoEs exhibit less superposition, we now turn to how these monosemantic features are organized within experts, and to what that organization reveals about specialization.

Conventional treatments of expert specialization in MoEs are framed around load balancing—the even distribution of inputs across experts (Chaudhari et al., 2025). This framing fails to capture the more natural intuition of specialization, under which an expert is invoked precisely when the concepts in which it specializes are present in the input, and not otherwise.

We therefore define an expert as specialized when it *occupies* certain feature directions in the input space and represents those features relatively monosemantically. We show that these two conditions are directly correlated, and that their joint presence itself promotes load balancing across experts—thereby recovering the conventional desideratum as a consequence, rather than a definition, of specialization. Specialization, on this account, is a statement about representational geometry, and even load distribution is one of its downstream effects.

Because we fix $k = 1$, the router partitions the feature space into convex cone regions (Appendix A.3), each region routed to a single expert. Formally, $\forall s > 0, \vec{x} \in C \rightarrow s\vec{x} \in C$, where C is the set of points contained in the cone and s is any positive scalar; that is, if a feature vector $\vec{x}$ is routed to an expert, then so is every positive rescaling $s\vec{x}$. The cone structure is what makes the notion of “occupancy” well defined: routing depends only on the direction of $\vec{x}$, not its magnitude. In this case we say that feature x is contained within expert e , and correspondingly that expert e *occupies* x .

We find empirically that small models which distribute the input space across more experts tend to achieve lower loss (Appendix A.4). This raises a natural question: does the alloca-

tion of the input space to particular experts entail any representational property of those experts? Our definition of specialization predicts that it entails *monosemanticity*—and, as we demonstrate below, this correlation is not confined to the smallest models but persists in larger toy models (e.g., $m = 10$).

For models with $n > 2$, we investigate the two complementary directions of this relationship. First, we ask whether initializing an expert to occupy specific features in the input space causes it to represent those features more monosemantically. Second, and conversely, we ask whether—for the features an expert has itself chosen to represent monosemantically—that expert in fact occupies those features in the input space. Confirming both directions would establish occupancy and monosemanticity as two faces of a single specialization phenomenon rather than as loosely associated tendencies.

Under diagonal initialization, each expert monosemantically represents exactly the single feature it initially occupied, and no other (Figure 5a). Under ordered k-hot initialization, the first expert monosemantically represents four of the five features it initially occupied (Figure 5b); the remaining experts, initialized over less important features, represent neither those less important features nor the five most important features they were not initialized over. When we break the ordering of feature importance and randomize the features each expert initially occupies, every expert monosemantically represents only the most important feature or features among those it was initialized over (Figure 5c).

The features initially routed to an expert are thus strongly correlated with the features that expert ultimately comes to represent monosemantically. Experts, moreover, represent *only* important features monosemantically—whether we initialize an expert with a single important feature explicitly, or supply it a set of features from which it selects the most important of its own accord and represents that one monosemantically. The first direction of the specialization relationship is therefore confirmed: occupancy at initialization determines which features an expert renders monosemantic.

Under uniform feature importance, by contrast, experts place all features in superposition with elevated levels of interference. Even in this regime, however, the features an expert initially occupies remain *relatively* more monosemantic, attaining superposition scores that lie, on average, 1.03 standard deviations below the mean for that expert—so that occupancy leaves a measurable imprint on representational geometry even when no feature is important enough to be represented monosemantically outright.

We next examined the converse correlation: whether an expert that represents certain features monosemantically also occupies those features in the input. To this end, we measure the expert’s usage both when its monosemantic features are *one of many* active features and when they are the *only* active features—the latter case being equivalent to measuring the probability that the expert occupies those features. The correlation holds robustly under both Xavier and k-hot initialization (Table 1). With $E = 10$, a mean expert usage of $\sim 10\%$ corresponds to even load balancing across experts. In every case, activating an expert’s monosemantic feature or features raises that expert’s usage substantially, and when those features are the sole active features the expert dominates routing outright. Under k-hot initialization the correspondence is exact: *100%* of the features an expert represents monosemantically are occupied by that same expert. Both directions of the relationship thus hold, establishing occupancy and monosemanticity as the two mutually entailing conditions of specialization we set out to define.

An expert that represents more features monosemantically is, by our definition, more specialized: its usage on arbitrary inputs stays near the balanced baseline, yet conditional on its specialized features being active its usage rises far above that of the remaining experts. This pattern holds in every case but one—the single Xavier-initialized expert representing four monosemantic features, whose overall utilization drops sharply—an isolated exception, supported by just one expert out of a thousand, set against an otherwise consistent monotonic trend spanning both initialization schemes. Specialization, so defined, therefore delivers exactly the behavior a well-functioning MoE is meant to exhibit: experts that lie dormant on irrelevant inputs and activate decisively on the inputs they are equipped to represent.

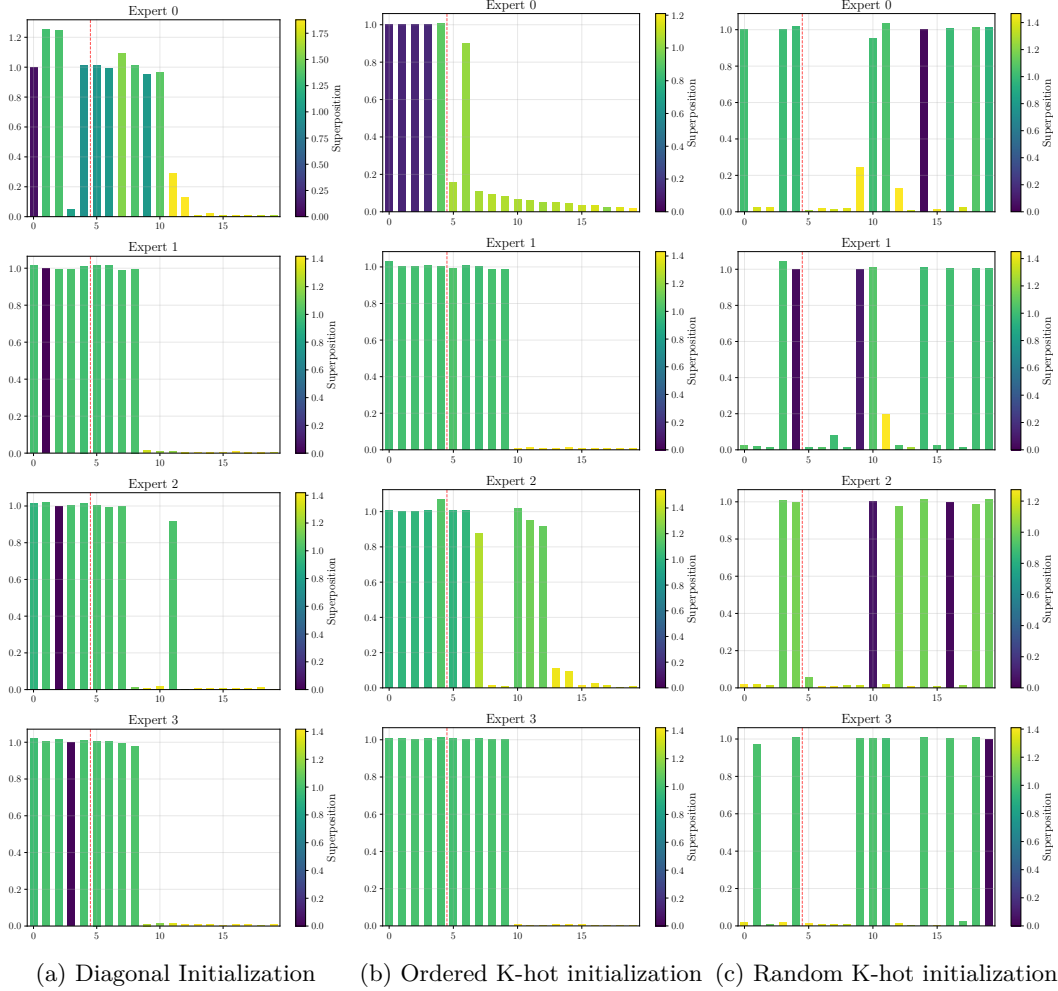


Figure 5: Expert feature norms $\|W_i^{(e)}\|$ and superposition (color) for three initialization schemes, with $n = 20, m = 5, E = 4, \hat{S} = 0.1$. In (a), the gate matrix is initialized along the main diagonal ($W_i^r = \hat{e}_i$, the basis vector for dimension i), and relative feature importance decays exponentially in order from feature one to twenty. In (b), the gate matrix is initialized to an “ordered k-hot” pattern in which the first expert aligns with the first five features and each subsequent expert with the next five; relative feature importance is as in (a). In (c), the gate matrix is initialized to a “random k-hot” pattern in which each expert is assigned five random features, with no feature shared between experts and all twenty features covered collectively; relative feature importance again decays exponentially but is now distributed randomly across features. Across all three schemes, the features an expert is initialized to occupy are exactly those it goes on to represent monosemantically, so long as they are among the more important features—the correspondence tightening as initialization aligns experts more directly with important features.

7 CONCLUSION

We investigated how experts shape superposition in MoEs and established three results. First, MoEs consistently exhibit greater monosemanticity than dense networks matched on parameters, and they forgo the discontinuous phase change altogether, transitioning ever more continuously as network sparsity increases. Second, network sparsity—not feature sparsity or feature importance—is the variable that governs this representational strategy. Third, expert specialization is more faithfully understood through monosemantic feature representation than through load balancing: on our feature-based definition, experts orga-

Table 1: Per-expert monosemantic-feature and usage statistics for $n = 100, m = 10, E = 10$. For each initialization scheme (Xavier and k-hot), one hundred models are trained, yielding 1000 experts per scheme. Statistics are aggregated across models, with experts grouped by the number of features they represent monosemantically. For the feature(s) an expert represents monosemantically, we report the expert’s usage when those features are one of several active features in the input and, separately, its usage when those features are the *only* active feature—the latter measuring how strongly the expert occupies those feature directions. Two patterns are consistent across both schemes: unconditional usage stays near the 10% even-load-balancing level regardless of how many features an expert renders monosemantic, whereas usage conditioned on those features being active rises sharply and monotonically with their number, reaching near-total dominance when those features alone are active (exactly 100% under k-hot initialization).

Xavier Initialization				
Number of monosemantic features per expert	Number of experts (out of 1000)	Mean expert usage (%)	Mean expert usage; feature(s) active (%)	Mean expert usage; only feature(s) active (%)
0	461	—	—	—
1	387	9.595	17.94	67.18
2	138	9.599	30.29	95.65
3	13	8.363	40.19	100.0
4	1	1.428	14.69	100.0
5	0	—	—	—
K-Hot Initialization				
Number of monosemantic features per expert	Number of experts (out of 1000)	Mean expert usage (%)	Mean expert usage; feature(s) active (%)	Mean expert usage only feature(s) active (%)
0	335	—	—	—
1	382	10.00	23.94	100.0
2	227	10.02	46.61	100.0
3	47	10.09	62.00	100.0
4	8	9.95	70.30	100.0
5	1	9.62	74.79	100.0

nize around coherent features whenever initialization encourages such specialization, and even load balancing emerges as a consequence of that organization. These findings rest on controlled autoencoder toy models with synthetic data, and their transfer to large-scale transformers—where the feature distribution is unknown—remains an open question (see Appendix A.6). Within these conditions, however, toy MoEs attain comparable reconstruction loss while sustaining markedly more interpretable representations, directly challenging the prevailing view that mechanistic interpretability and model capability stand in fundamental tension. Future work should determine what favors monosemanticity in MoEs, how their training dynamics diverge from those of dense models, and when specialization emerges; resolving these questions would directly inform the design of more interpretable and high-performing language models.

REFERENCES

Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermyn, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, Robert Lasenby, Yifan Wu, Shauna Kravec, Nicholas Schiefer, Tim Maxwell, Nicholas Joseph, Zac Hatfield-Dodds, Alex Tamkin, Karina Nguyen, Brayden McLean, Josiah E Burke, Tristan Hume, Shan Carter, Tom Henighan, and Christopher Olah. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023. <https://transformer-circuits.pub/2023/monosemantic-features/index.html>.

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- Marmik Chaudhari, Idhant Gulati, Nishkal Hundia, Pranav Karra, and Shivam Raval. Moe lens - an expert is all you need. In *Sparsity in LLMs (SLLM): Deep Dive into Mixture of Experts, Quantization, Hardware, and Inference*, 2025. URL <https://openreview.net/forum?id=GS4WXncwSF>.
- Michael B. Cohen, Yin Tat Lee, Cameron Musco, Christopher Musco, Richard Peng, and Aaron Sidford. Uniform sampling for matrix approximation. *CoRR*, abs/1408.5099, 2014. URL <http://arxiv.org/abs/1408.5099>.
- Nishanth Dikkala, Nikhil Ghosh, Raghu Meka, Rina Panigrahy, Nikhil Vyas, and Xin Wang. On the benefits of learning to route in mixture-of-experts models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 9376–9396, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.583. URL <https://aclanthology.org/2023.emnlp-main.583>.
- Nelson Elhage, Tristan Hume, Catherine Olsson, Nicholas Schiefer, Tom Henighan, Shauna Kravec, Zac Hatfield-Dodds, Robert Lasenby, Dawn Drain, Carol Chen, Roger Grosse, Sam McCandlish, Jared Kaplan, Dario Amodei, Martin Wattenberg, and Christopher Olah. Toy models of superposition. *Transformer Circuits Pub*, 2022. URL https://transformer-circuits.pub/2022/toy_model/index.html.
- William Fedus, Barret Zoph, and Noam Shazeer. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity, 2022. URL <https://arxiv.org/abs/2101.03961>.
- Google DeepMind. Gemini 2.5 pro. <https://deepmind.google/technologies/gemini/pro/>, 2025.
- Wes Gurnee, Neel Nanda, Matthew Pauly, Katherine Harvey, Dmitrii Troitskii, and Dimitris Bertsimas. Finding neurons in a haystack: Case studies with sparse probing. *arXiv preprint arXiv:2305.01610*, 2023.
- Albert Q Jiang, Alexandre Sablayrolles, Antoine Roux, Arthur Mensch, Blanche Savary, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Emma Bou Hanna, Florian Bressand, et al. Mixtral of experts. *arXiv preprint arXiv:2401.04088*, 2024.
- Jakub Krajewski, Jan Ludziejewski, Kamil Adamczewski, Maciej Pióro, Michał Krutul, Szymon Antoniak, Kamil Ciebiera, Krystian Król, Tomasz Odrzygóźdź, Piotr Sankowski, Marek Cygan, and Sebastian Jaszczur. Scaling laws for fine-grained mixture of experts. *arXiv preprint arXiv:2402.07871*, 2024. URL <https://arxiv.org/abs/2402.07871>.
- Dmitry Lepikhin, HyukJoong Lee, Yuanzhong Xu, Dehao Chen, Orhan Firat, Yanping Huang, Maxim Krikun, Noam Shazeer, Zhifeng Chen, Yonghui Wu, et al. Gshard: Scaling giant models with conditional computation and automatic sharding. In *Proceedings of the 38th International Conference on Machine Learning*. PMLR, 2021.
- Siyuan Mu and Sen Lin. A comprehensive survey of mixture-of-experts: Algorithms, theory, and applications, 03 2025.
- Jungwoo Park, Ahn Young Jin, Kee-Eung Kim, and Jaewoo Kang. Monet: Mixture of monosemantic experts for transformers. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=10gw1SHY3p>.
- Kiho Park, Yo Joong Choe, and Victor Veitch. The linear representation hypothesis and the geometry of large language models, 2024. URL <https://arxiv.org/abs/2311.03658>.
- Lucas Prieto, Edward Stevinson, Melih Barsbey, Tolga Birdal, and Pedro AM Mediano. Correlations in the data lead to semantically rich feature geometry under superposition. In *Mechanistic Interpretability Workshop at NeurIPS 2025*, 2025.
- Adam Scherlis, Kshitij Sachan, Adam S. Jermyn, Joe Benton, and Buck Shlegeris. Polysemanticity and capacity in neural networks, 2025. URL <https://arxiv.org/abs/2210.01892>.

-
- Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarczyk, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer, 2017. URL <https://arxiv.org/abs/1701.06538>.
- An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025a.
- Xingyi Yang, Constantin Venhoeff, Ashkan Khakzar, Christian Schroeder de Witt, Puneet K. Dokania, Adel Bibi, and Philip Torr. Mixture of experts made intrinsically interpretable. In *Forty-second International Conference on Machine Learning*, 2025b. URL <https://openreview.net/forum?id=6QERrXMLP2>.
- Yanqi Zhou, Tao Lei, Hanxiao Liu, Nan Du, Yanping Huang, Vincent Y Zhao, Andrew M. Dai, Zhifeng Chen, Quoc V Le, and James Laudon. Mixture-of-experts with expert choice routing. In Alice H. Oh, Alekh Agarwal, Danielle Belgrave, and Kyunghyun Cho (eds.), *Advances in Neural Information Processing Systems*, 2022. URL <https://openreview.net/forum?id=jdJo1HIVinI>.
- Barret Zoph, Irwan Bello, Sameer Kumar, Nan Du, Yanping Huang, Jeff Dean, Noam M. Shazeer, and William Fedus. St-moe: Designing stable and transferable sparse expert models, 2022. URL <https://api.semanticscholar.org/CorpusID:248496391>.

A APPENDIX

A.1 MEASURING LOSS FOR VARYING SPARSITY & EXPERTS

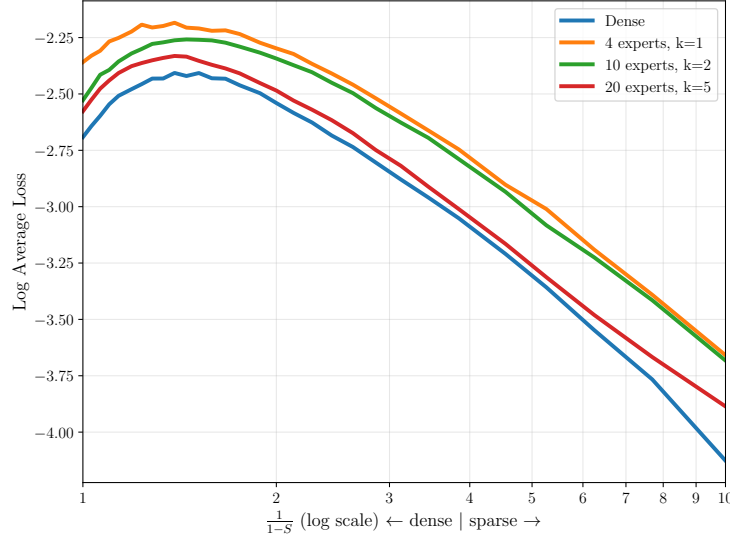


Figure 6: Log average loss versus feature density ($\frac{1}{1-s}$) for dense ($m = 20$) and MoE (4 experts, $k = 1$, $m = 5$), MoE (10 experts, $k = 2$, $m = 2$), and MoE (20 experts, $k = 5$, $m = 1$) models, all with uniform feature importance ($I_i = 1.0$) for $n = 100$ input features. Results are averaged over five runs per sparsity level. The dense model attains the lowest loss at every sparsity level, but the margin over the MoEs is small and shrinks monotonically as the expert count increases: the MoE loss converges toward that of the dense model, quantifying the modest and diminishing price paid for the substantially greater monosemanticity documented in Figure 3.

A.2 THEORETICAL INTUITION FOR SUPERPOSITION

In dense models, superposition emerges to exploit the gap between sparse features and dense computation, compressing rarely active features into shared dimensions. MoEs, by contrast, close this same gap structurally, through conditional computation: by aligning activation sparsity with feature sparsity, they eliminate the computational penalty that dedicated, rarely activating neurons would otherwise incur. As Elhage et al. (2022) observe, once a model expends computation only on active features, splitting polysemantic neurons into dedicated monosemantic ones becomes the optimal strategy—in effect trading the *compression* of superposition for the *selection* afforded by routing. Conditional computation thus removes the very incentive that drives dense models into superposition.

This structural change manifests geometrically as a reduction in *interference*. Although the global ratio of represented features to parameters is unchanged, the router partitions the feature space, so that a feature routed to expert e competes for capacity only with the subset of features also assigned to that expert. Interference is consequently governed by the local expert matrix $(W^e)^\top W^e$ rather than by the global $W^\top W$ of a dense model. This partitioning sharply reduces the number of interfering features for any given feature vector, relieving the optimization pressure to pack features in superposition and thereby permitting experts to learn monosemantic representations—the mechanistic account of the empirical monotonic trend reported in Section 4.

A.3 ROUTER SUBSPACES ARE CONVEX CONES

In the regime $k = 1$, the router function $g(x) = \text{softmax}(W^r x)$ is equivalent to $\text{argmax}(W^r x)$, since the softmax preserves the ordering of its inputs. The region routed to expert i can then be written as $\forall j \neq i, (w_i - w_j)^T x > 0$, where w_i and w_j are row vectors of W^r . Each such constraint is a homogeneous linear inequality, and the intersection of finitely many homogeneous linear inequalities is by definition a convex cone: if a particular x satisfies every inequality, then so does sx for any positive scalar s ; and if x_1 and x_2 both satisfy them, then so does every convex combination $x = \lambda x_1 + (1 - \lambda)x_2$ for $\lambda \in [0, 1]$. The routing region of each expert is therefore a convex cone, which is exactly what licenses the direction-based notion of occupancy used in Section 6.

When $k > 1$, the region of inputs routed to a particular expert e becomes a union of such convex cones. Because a union of convex cones is not in general itself a convex cone, the characterization of experts as occupying feature directions is guaranteed only for $k = 1$, and we accordingly restrict that characterization to this regime.

A.4 EXPERT ROUTING WITH DIFFERENT INITIALIZATION SCHEMES

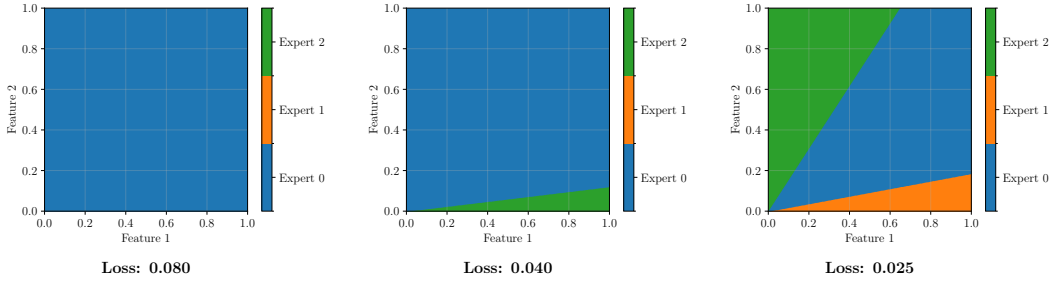


Figure 7: Expert routing of three identical models under differing initialization schemes, with $n = 2$, $E = 3$, $m = 1$. The three models trace a clear monotonic relationship between how widely the input space is distributed across experts and reconstruction loss. The first model (left) attains the worst performance (loss: 0.08) and routes all inputs to a single expert. The second model (middle) performs better (loss: 0.04), routing a small portion of inputs—specifically those for which feature 1 is active—to a second expert. The third model (right) achieves the lowest loss (loss: 0.025) and distributes the input space across all experts: one expert is chosen when only feature 1 is active, one when only feature 2 is active, and one when both are active.

In small models ($n = 2$, $m = 1$, $E > 1$), sweeping $E \in [2, 7]$, we find empirically that models distributing the input space across more experts tend to achieve lower loss. Fixing $n = 2$ lets us visualize directly which portions of the input space are routed to which experts (Figure 7), making the connection between broad input-space coverage and low loss immediately legible.

A.5 ANALYTIC MODEL EQUIVARIANCE

For the toy setup of single-layer, single-nonlinearity, top- $k = 1$ MoEs, there exists a theoretical map from any dense model to a monosemantic MoE that represents an equivalent number of active features under a sparsity constraint.

Assume there exists an upper bound a on the number of active features for any input, so that $\forall x \in D : |\{i : x_i \neq 0\}| \leq a$. Assume further that a is no greater than the hidden dimensionality m of an expert—which bounds the number of features the model must represent—and that this hidden dimensionality is in turn no larger than the total number of input features n (i.e. $a \leq m \leq n$). To construct the monosemantic MoE, for each subset $S \subseteq \{1, 2, \dots, n\}$ with $|S| \leq a$ —that is, each subset of active features of size at most a —we instantiate an expert that monosemantically preserves those features. (It in fact suffices to take only the subsets with $|S| = a$.) The router then selects the expert corresponding to the active features, of which, by assumption, there are never more than a :

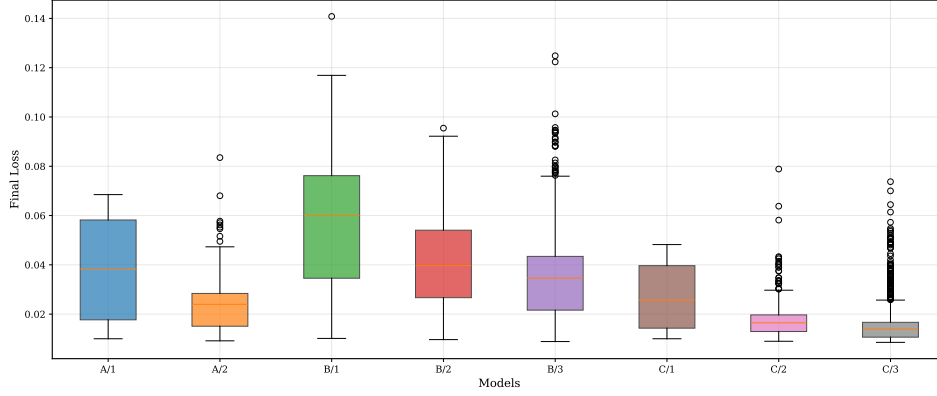


Figure 8: Model X/E uses X to denote the same model architectures as Figure 4 and E to denote the total number of experts (i.e. network sparsity). Increasing network sparsity lowers mean loss while raising localized variance—the effect strengthening as the expert count approaches the number of input feature dimensions. We attribute this variance to the relatively less stable training of MoEs compared with dense models, which persists despite our training ten models per cell and retaining only the lowest-loss run.

$$\text{Router}(x) = \arg \max_S \mathbb{I}[\text{support}(x) = S]$$

where $\text{support}(x) = \{i : x_i \neq 0\}$. Since $|S| \leq a \leq m$, each expert possesses sufficient capacity to represent its assigned features without superposition. To restate the construction: at most a features are ever active, every distinct combination of active features receives its own dedicated expert, and each such expert has capacity sufficient to represent those features monosemantically. The number of experts required is therefore $\binom{n}{m}$.

The reconstruction of this theoretical MoE attains zero loss precisely insofar as the sparsity constraint holds (equivalently, as sparsity approaches one in these toy models), since more than m features could in principle be active simultaneously ($a \not\leq m$), which would exceed the monosemantic representational capacity of the network. Even then, however, the dense polysemantic model could do no better unless features are correlated in the distribution: whenever $a \not\leq m$, the polysemantic model confronts the identical obstacle, so the monosemantic MoE under this construction may still match or outperform it under looser sparsity constraints.

Thus, for any dense model $f_{\text{dense}}(x) = \text{ReLU}(Wx + b)$ operating under the sparsity constraint $|\text{support}(x)| \leq a$, there exists an MoE model $f_{\text{MoE}}(x)$ with $f_{\text{dense}}(x) = f_{\text{MoE}}(x)$ for all valid inputs. In the toy settings of this paper, the sparsity constraint holds in the limit where sparsity approaches one. In practice, moreover, the number of features that a given quantity of information can semantically encode may itself be bounded—as the dimensionality of meaningful embeddings of that data suggests—so that an MoE with sufficiently many experts and a tractable, interpretable degree of superposition may suffice to encode every feature present.

A.6 LIMITATIONS

Our results should be read in light of several limitations that also delineate the scope of our claims. First, all experiments are conducted in controlled toy-model settings derived from superposition studies in sparse autoencoders (Elhage et al., 2022). This design affords precise measurement of interference and monosemanticity—the very control that makes our attributions to network sparsity credible—but it abstracts away many complexities of large-scale transformers, including multiple layers of computation, attention, heterogeneous feature distributions, and realistic routing dynamics; the transferability of our findings to production MoE architectures thus remains to be established.

Second, our architectural and routing choices are deliberately restricted: simple experts, fixed top- k routing (most often $k = 1$), equal parameter budgets between dense and MoE models, and no auxiliary load-balancing losses. Practical MoEs frequently employ richer routing mechanisms, variable expert capacities, and multi-task objectives (Lepikhin et al., 2021; Fedus et al., 2022), or extend beyond reconstruction to objectives such as next-token prediction—any of which may yield qualitatively different representational behavior.

Third, the feature distributions of our synthetic tasks—their sparsity patterns, feature-importance profiles, and independence assumptions—are considerably simpler than those encountered in natural data.

Finally, although we observe increased monosemanticity and reduced superposition in MoEs under these fixed conditions, we do not evaluate downstream performance trade-offs, training stability, or specialization dynamics at scale; prior work indicates that such dynamics can shift substantially with model size, optimization regime, and data diversity (Krajewski et al., 2024).

In sum, our study delivers mechanistic insight under clean, controlled conditions, and validating these patterns in large, realistic MoE systems is the natural next step. Because our present methods do not scale directly to such models, developing scalable measurements of superposition is itself a clear and compelling direction for future work.